Trigonometry

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Contents

1 Ray **3**

2 Angle **3**
2.1 Standard Position 3
2.2 Measurement 3
2.3 Angle Type Names 3
2.4 Units 4

3 Radian Measurement **4**
3.1 Definition of a Radian 4
3.2 Angle Measurement 4
3.3 Review 5
3.4 Definition of Radian Angle Measure 5
3.5 Converting Degrees to Radians 5

4 Trigonometric Functions **5**
4.1 The Unit Circle 5
4.2 Functions 6
4.3 Functions of Real Numbers and Angles 6

5 Using Trigonometric Functions to Determine Point Values on a Unit Circle **6**

6 Fundamental Trigonometric Identities **7**

7 Pythagorean Identities **7**

8 Periodic Functions **7**
8.1 Periodic Properties of Trigonometric Functions 8

9 Even and Odd Properties of Trigonometric Functions **8**

10 Trigonometric Functions of Any Angle **9**

11 Reference Angles **9**
11.1 Definition 9
11.2 Evaluate Trigonometric Functions using Reference Angles 10

12 Right Triangle Trigonometry 4.3 **10**
12.1 Anatomy of a Triangle 10
12.2 Definition of Trigonometric Functions of Acute Angles 10
12.3 Pythagorean Theorem 11
12.4 Cofunction Identities 11
12.5 Angles of Elevation and Depression 11

13 Graphs of Trigonometric Functions **11**
13.1 Graphing y = sine x and y = cos x 11
13.2 Trigonometric Function Transformations 12
13.3 Properties of General Sine and Cosine Functions 12
13.4 Producing a Function According to Given Specifications 13
13.5 Graphing Cosecant Functions 13
13.6 Graphing Secant Functions 14
13.7 Graphing Tangent Functions 14
13.8 Universal Transformation Formulas 14

14 Triangles **14**
14.1 Types Of Triangles 14
14.2 Special Triangles 15

15 Inverse Trigonometric Functions **15**
15.1 Inverse Sine Function or Arcsin 15
15.2 Inverse Cosine Function or Arccosine 16
15.3 Inverse Tangent Function 16
15.4 Inverse Secant Function 17
15.5 Inverse Cosecant Function 17
15.6 Inverse Cotangent Function 17
15.7 Inverse Function Use Cases 17

16 5.1: Algebraic manipulation of Trigonometric Functions **18**
16.1 LCD of Trigonometric Functions and Variables 18
16.2 Properties of Trigonometric Functions 18
16.3 Proving Trigonometric Identities 18

17 5.2: Sum and Difference Formulas **19**
17.1 Chord 19
17.2 Sine Sum Formula 19
17.3 Sine Difference Formula 19
17.4 Cosine Sum Formula 19
17.5 Cosine Difference Formula 19
17.6 Tangent Sum Formula 19

17.7 Tangent Difference Formula 19

18 5.3: Double-Angle, Power-Reducing, and Half-Angle Measurements 20

18.1 Double-Angle Formulas 20

18.2 Power-Reducing Formulas 20

18.3 Half-Angle Formulas 20

1 Ray

A ray is part of a line that consists of an endpoint and all points on the line to one side of the endpoint.

A ray is denoted symbolically by $\overrightarrow{PQ}$ where P is the endpoint and Q is another point on the ray to the right of P. A ray always has one endpoint which is always written first when naming the ray.

2 Angle

An angle is formed by rotating a ray about it's endpoint. The starting position is called the initial side of the angle, and the final position of the ray is called the terminal side. The common endpoint is called the vertex of the angle, and the vertex is often denoted by a capitlal letter such as A.

An angle is notated by $\angle A$ or by $\angle BAC$, where B is the initial side, A being the vertex, and C being the terminal side.

2.1 Standard Position

An angle is said to be at standard position if it's vertex is at the origin in the xy-plane, and the initial ray is directly on the x axis.

2.2 Measurement

The measure of an angle quantifies the direction and amount of rotation from the initial ray to the terminal ray. The measure is positive if the rotation is counterclockwise, and the measurement is negative if the rotation is clockwise. Often, degrees are used to measure the amount of rotation, one full rotation of a ray about its vertex is equivalent to 360°

2.3 Angle Type Names

1. Right Angle:
   Right angles are angles at 90° measure.

2. Straight Angle:
   Straight angles are angles at 180° measure.

3. Acute Angle:
   Acute angles are angles at $0° < \theta < 90°$

4. Obtuse Angle:
   Obtuse angles are angles at $90° < \theta < 180°$

5. Complementary Angles:
   A pair of angles which sum to 90°

6. Supplementary Angles:
   A pair of angles which sum to 180°

7. Coterminal Angles:
   Two angles in standard position with the same initial side and same terminal side are called coterminal angles.

2.4 Units

1. Minutes / Arcminutes (min or $'$):
   A singular degree can be divided into 60 equal parts; each 1/60th is referred to as one arcminute / minute. $\left(\frac{1}{60}\right)^{\circ} = 1' \vee 1min$

2. Seconds / Arcseconds (sec or "):
   A singular arcminute can be divided into 60 equal parts; each 1/60th of such is referred to as one arcsecond. $\left(\frac{1}{60}\right)^{'} = 1'' \vee 1sec$

3 Radian Measurement

3.1 Definition of a Radian

A central angle that incercepts an arc on the circle with length equal to the radius of the circle has a measure of one radian.

3.2 Angle Measurement

When two lines or rays cross a circle, the part of the circle between the intersection points is called the intercepted arc and is often denoted by s.

Any central angle can be measured in radians by dividing the length s of the intercepted arc by the radius r.

3.3 Review

1. π:
   is defined as the ratio of the circumference of a circle to its diameter d.

2. Circumference:
   Therefore the circumference is given by $C = \pi d$ or $C = 2\pi r$, where r is the radius of the circle, dividing this by the radius gives the number of radians in one revolution. The angular measure of one full rotation is 2π

3.4 Definition of Radian Angle Measure

The radian measure of a central angle θ subtended by an arc of length s on a circle of radius r is given by $\theta = \frac{s}{r}$. radian measure carries no units because it is measured as a ratio of two lengths with the same units (the units associated with s/r "cancel"). Thus, 2π radians is simply written as 2π. It is universally understood that the measure is in radians. Sometimes the notation "rad" is included for emphasis, but is not necessary.

1. 1 revolution: $2\pi = 360°$
2. $\frac{1}{2}$ revolution: $\pi = 180°$
3. $\frac{1}{4}$ revolution: $\frac{\pi}{2} = 90°$
4. $\frac{3}{4}$ revolution: $\frac{3\pi}{2} = 270°$

3.5 Converting Degrees to Radians

To convert to radians from degrees multiply the degree by $\frac{\pi}{180°}$

$$x^\circ \cdot \frac{\pi}{180^\circ} = x_{rad}$$

To convert from radians to degrees multiply the degree by $\frac{180°}{\pi}$

$$x_{rad} \cdot \frac{180^\circ}{\pi} = x^\circ$$

4 Trigonometric Functions

4.1 The Unit Circle

The unit circle consists of all points (x,y) that satisfy the equation $x^2 + y^2 = 1$

4.2 Functions

Let $P(x, y)$ be the point associated with a real number t measured along the circumference of the unit circle from the point (1,0).

1. sine:
   $\sin t = y$
2. cosine:
   $\cos t = x$
3. tangent:
   $\tan t = \frac{y}{x} (x \neq 0)$
4. cosecant:
   $\csc t = \frac{1}{y} (y \neq 0)$
5. secant:
   $\sec t = \frac{1}{x} (x \neq 0)$
6. cotangent:
   $\cot t = \frac{x}{y} (y \neq 0)$

If P is on the y-axis in a unit circle example, then x = 0 and the tangent and secant functions are undefined.

If P is on the x-axis in a unit circle example, then y = 0 and the cotangent and cosecant functions are undefined.

4.3 Functions of Real Numbers and Angles

If $\theta = t$ rad, then:

$$\sin t = \sin \theta$$
$$\cos t = \cos \theta$$
$$\tan t = \tan \theta$$
$$\csc t = \csc \theta$$
$$\cot t = \cot \theta$$
$$\sec t = \sec \theta$$

5 Using Trigonometric Functions to Determine Point Values on a Unit Circle

In a unit circle where the *radius* = 1 you can denote the equation of the circle using $x^2 + y^2 = r$, this means you can plug in $\sin t$'s or $\cos t$'s t value into the circle equation to produce the other value and thus the coordinate set for point locations, you will have to have a qualifier for the quadrant within which the real coordinate lies in order to determine the real $P(x, y)$ value.

6 Fundamental Trigonometric Identities

When computing the value of a trigonometric function the argument MUST be included.

1. $\sin t$ and $\csc t$ are reciprocals

$$\csc t = \frac{1}{\sin t} \vee \sin t = \frac{1}{\csc t}$$

2. $\cos t$ and $\sec t$ are reciprocals

$$\sec t = \frac{1}{\cos t} \vee \cos t = \frac{1}{\sec t}$$

3. $\tan t$ and $\cot t$ are reciprocals

$$\cot t = \frac{1}{\tan t} \vee \tan t = \frac{1}{\cot t}$$

4. $\tan t$ is the ratio of $\sin t$ and $\cos t$.

$$\tan t = \frac{\sin t}{\cos t}$$

5. $\cot t$ is the ratio of $\cos t$ and $\sin t$

$$\cot t = \frac{\cos t}{\sin t}$$

7 Pythagorean Identities

1. $$\sin^2 t + \cos^2 t = 1$$

2. $$\tan^2 t + 1 = \sec^2 t$$

3. $$1 + \cot^2 t = \csc^2 t$$

8 Periodic Functions

A function f is periodic if $f(t+p) = f(t)$ for some constant p.
The smallest positive value p for which f is periodic is called the period of f.

8.1 Periodic Properties of Trigonometric Functions

The values of the six trigonometric functions of t are determined by the corresponding point $P(x, y)$ on the unit circle. The Circumference of the unit circle is 2π, adding (or subtracting) 2π to t results in the same terminal point (x, y). Consequently, the values of the trigonometric functions are the same for t and $t + 2n\pi$

1. Sine:
   $period = 2\pi$
   $property = \sin(t + 2\pi) = \sin t$

2. Cosine:
   $period = 2\pi$
   $property = \cos(t + 2\pi) = \cos t$

3. Cosecant:
   $period = 2\pi$
   $property = \csc(t + 2\pi) = \csc t$

4. Secant:
   $period = 2\pi$
   $property = \sec(t + 2\pi) = \sec t$

5. Tangent:
   $period = \pi$
   $property = \tan(t + \pi) = \tan t$

6. Cotangent:
   $period = \pi$
   $property = \cot(t + \pi) = \cot t$

If the period is p, then $f(t + p) = f(t)$. It is also true that $f(t + np) = f(t)$ for any integer n. That is, adding any integer multiple of the period to a domain element of a periodic function results in the same function value.

9 Even and Odd Properties of Trigonometric Functions

1. Sine:
   $\sin t = y$ and $\sin(-t) = -y$
   Odd Function: $\sin(-t) = -\sin t$

2. Cosine:
   $\cos t = x$ and $\cos(-t) = x$
   Even Function: $\cos(-t) = \cos t$

3. Cosecant:
   $\csc t = \frac{1}{y}$ and $\csc(-t) = \frac{1}{-y}$
   Odd Function: $\csc(-t) = -\csc t$

4. Secant:
   $\sec t = \frac{1}{x}$ and $\sec(-t) = \frac{1}{x}$
   Even Function: $\sec(-t) = \sec t$

5. Tangent:
   $\tan t = \frac{y}{x}$ and $\tan(-t) = \frac{-y}{x}$
   Odd Function: $\tan(-t) = -\tan t$

6. Cotangent:
   $\cot t = \frac{x}{y}$ and $\cot(-t) = \frac{x}{-y}$
   Odd Function: $\cot(-t) = -\cot t$

10 Trigonometric Functions of Any Angle

Let θ be an angle in standard position with point $P(x, y)$ on the terminal side, and let $r = \sqrt{x^2 + y^2} \neq 0$ represent the distance from $P(x, y)$ to $(0, 0)$. Then:

$$\sin\theta = \frac{y}{r}$$
$$\cos\theta = \frac{x}{r}$$
$$\tan\theta = \frac{y}{x}(x \neq 0)$$
$$\csc\theta = \frac{r}{y}(y \neq 0)$$
$$\sec\theta = \frac{r}{x}(x \neq 0)$$
$$\cot\theta = \frac{x}{y}(y \neq 0)$$

11 Reference Angles

11.1 Definition

Let θ be an angle in standard position. The reference angle for θ is the acute angle θ' formed by the terminal side of θ and the horizontal axis.
The length of $\angle\theta'$s vertical leg is $|y|$.
The length of the horizontal leg is $|x|$.
The hypotenuse's length is $r = \sqrt{x^2 + y^2}$

11.2 Evaluate Trigonometric Functions using Reference Angles

$\angle\theta$ = standard position
$\angle\theta'$ = reference angle complementary to $\angle\theta$

$$\cos\theta = \frac{x}{r} \wedge \cos\theta' = \frac{adj}{hyp} = \frac{|x|}{r}$$

To Find the value of a trigonometric function of a given angle θ

1. Determine the function value of the reference angle θ'
2. Affix the appropriate sign based on the quadrant in which θ lies

12 Right Triangle Trigonometry 4.3

12.1 Anatomy of a Triangle

Consider a right triangle, the longest side of the triangle is called the hypotenuse, the remaining two legs are determined relative to the position of the considered angle. There is an acute angle θ, the leg opposite of this angle is considered the opposite leg, the adjacent leg is, similarly, considered the adjacent leg.

The six trigonometric functions take inputs relative to the acute angle θ to determine the ratios of the lengths of the sides of the triangle.

12.2 Definition of Trigonometric Functions of Acute Angles

$$sine = \sin\theta = \frac{opp}{hyp}$$

$$cosine = \cos\theta = \frac{adj}{hyp}$$

$$tangent = \tan\theta = \frac{opp}{adj}$$

$$cosecant = \csc\theta = \frac{hyp}{opp}$$

$$secant = \sec\theta = \frac{hyp}{adj}$$

$$cotangent = \cot\theta = \frac{adj}{opp}$$

The mnemonic device "SOH-CAH-TOA" may help you remember the ratios for $\sin\theta$, $\cos\theta$, $\tan\theta$, respectively.

12.3 Pythagorean Theorem

legs a, b;
hypotenuse c;

$$a^2 + b^2 = c^2$$

12.4 Cofunction Identities

Cofunctions of complementary angles are equal; the "co" prefixing any one of these functions stands for "complementary".

1. Sine and Cosine are Cofunctions:

$$\sin\theta = \cos(90^\circ - \theta)$$
$$\cos\theta = \sin(90^\circ - \theta)$$

2. Tangent and Cotangent are Cofunctions:

$$\tan\theta = \cot(90^\circ - \theta)$$
$$\cot\theta = \tan(90^\circ - \theta)$$

3. Secant and Cosecant are Cofunctions:

$$\sec\theta = \csc(90^\circ - \theta)$$
$$\csc\theta = \sec(90^\circ - \theta)$$

12.5 Angles of Elevation and Depression

An angle of elevation gives the measure upwards from a horizontal line of reference.
An angle of depression gives the measure downwards from a horizontal line of reference.

cos X sin X csc X sec X tan X cot X

13 Graphs of Trigonometric Functions

13.1 Graphing y = sine x and y = cos x

Recall that $\sin\theta = \frac{y}{r}$, thus if you measure $y = \sin x$ you will produce a wave about the x axis with a peak of 1 and a trough of -1 which resembles a semi-circle alternating across the $\pm$y-plane as it approaches $\pm$ inf on x-plane.

The cos function performs similarly being that they are cofunctions, however they differ about the origin in that sin which represents the $\frac{y}{r}$ value for a unit circle example will start at 0 whereas cos, being that it starts at $\frac{x}{r}$ will start at 1

13.2 Trigonometric Function Transformations

Similarly to other functions, $\sin \wedge \cos$ can have function transformations performed upon their parent function to produce predictably altered graphs.
Phase Shift is another word for the horizontal translation of a trigonometric function.
Horizontal scaling causes period changes because it stretches or compresses the amount inputed into the parent function.
The function transformations seem to apply generically to trigonometric functions in a manner similar to basic algebraic functions.

13.3 Properties of General Sine and Cosine Functions

The Sine wave form starts halfway between it's crest and trough because the degree input 0 corresponds to point P(1,0) on the unit circle.

The Cosine wave form starts at the crest of it's wave because it represents the x-value on a unit circle of the degree inputed, i.e. $0 = 1$ because degree zero corresponds to the point P(1,0) on the unit circle.

Recall transformation format:

$$a \cdot f(b(x-h)) + k$$

Sine and Cosine functions measure period using the following format for convenience:

$$y = A \cdot f(Bx - C) + D$$

This is just a distributed form of the above, where the horizontal scaler has been distributed over the parent functions inner quantity.

Taking the above into account, you can easily determine the horizontal scaler and phase shift (horizontal translater) by reverse distributing the coefficient of the x-value, B.

In the following explanations I will be using the variable names from both of the above forms to refer to measurements.

1. Amplitude is determined by taking the absolute value of the vertical scaler variable $|a| \vee |A|$.

2. The period is determined by:

$$\frac{2\pi}{B}(B > 0)$$

   If $B < 0$ then rewrite the function using the odd and even properties before evaluation.

3. The phase shift is determined by h or:

$$\frac{C}{B}$$

4. The vertical shift is determined by D or k.

13.4 Producing a Function According to Given Specifications

Where crest is the highpoint and trough is the lowpoint.

$$x = crest$$

$$y = trough$$

1. Amplitude is found by halving the distance between the highest and lowest values.
2. Cosine Period: Twice the distance from the first maximum value to the minimum value.
3. Cosine Vertical Shift: The midpoint of the range from the highest to lowest value gives the vertical shift value:

$$-|A| + D \leq y \leq |A| + D$$

4. Phase Shift: Where the graph beings = y. find using normal $\frac{C}{B} = y$ algebraic deduction.

13.5 Graphing Cosecant Functions

In the cosecant function,

$$\frac{r}{y} \vee \frac{1}{\sin\theta}$$

a vertical asymptote appears with a period of π, because when the degree lies on the x-axis y is zero. Therefore the function,

$$f(x) = \csc x$$

Will have vertical asympotes at,

$$f(\pi), f(2\pi), f(3\pi), ...$$

and so on.

The relative minima and maxima of the sine function correspond to the relative minima and maxima of the base of the quadratic like in shape, functions which occur between the vertical asympotes.

13.6 Graphing Secant Functions

The Secant function, corresponding to the cosine function as it's reciprocal, is rather like the cosecant function however it's period of vertical asymptote production is different, occuring at every $\frac{\pi}{2}$ mark, i.e. along the y-axis. This is the main differentiator between the Secant function and the Cosecant function.

13.7 Graphing Tangent Functions

Recall that the $\tan\theta$ is an odd function. It's period is π. The function resembles a cube root function which repeats itself between vertical asympotes which lie along the x-axis and y-axis in terms of x and y unit circle measurements.

13.8 Universal Transformation Formulas

Transformation Format:

$$y = A \cdot f(Bx - C) + D$$

1. Amplitude:

$$Amp = |A|$$

2. Phase Shift:

$$PS = C$$

3. Vertical Translation:

$$VT = D$$

4. Period:

   $period_{parent}$ = the parent functions period

$$P = \frac{period_{parent}}{|B|}$$

14 Triangles

The total of all three angles in a triangle is always 180°.

14.1 Types Of Triangles

1. Equilateral Triangle:
   All three sides are equal in length, and all three angles are 60°.

2. Isosceles Triangle:
   At least two sides are equal in length; the angles opposite the equal sides are also equal.

3. Scalene Triange:
   All three sides have different lengths, and all three angles are different.

4. Acute Triangle:
   All three angles are less than 90°

5. Right Triangle:
   One angle is exactly 90°; the other two are acute.

6. Obtuse Triangle:
   One angle > 90°, all sides different.

14.2 Special Triangles

Isosceles Right Triangles (Angle Measurements: $45° - 45° - 90°$)
The hypotenuse is equivalent to either of the legs multiplied by the square root of two. For legs a and b, and hypotenuse c:

$$a = b \wedge c = a\sqrt{2} \vee b\sqrt{2}$$

Scalene Right Triangle (Angle Measurements: $30° - 60° - 90°$)
The side opposite 30° is the shortest. The side opposite 60° is the longer leg. Where the shortest leg $= a$, the longer leg $= b$, and the hypotenuse $= c$:

$$b = a\sqrt{3}$$

$$c = 2x$$

15 Inverse Trigonometric Functions

15.1 Inverse Sine Function or Arcsin

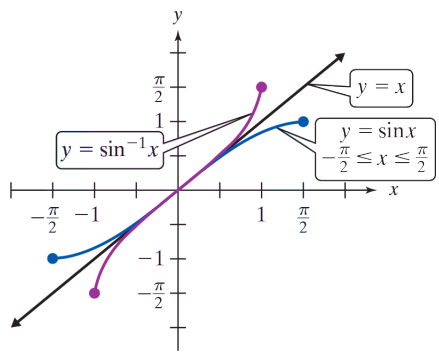
In it's base form the function $y = \sin x$ is only injective if it is restricted to a singular trough to crest half wave form.

$$y = \sin x \iff -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$

Restrictions Caused on $y = \sin^{-1} x$:

$$-1 \leq x \leq 1 \wedge -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

The inverse of the restricted sine function is called arcsin or $\sin^{-} 1$ in traditional inverse function denotation.



15.2 Inverse Cosine Function or Arccosine

Arccosine is similarly restricted although over a different domain of x values for it's parent function given that cosine starts at one by the origin, not zero, and is an even function.

Restrictions for $y = \cos x$:

$$0 \leq x \leq \pi$$

$$-1 \leq y \leq 1$$

$$y = \cos^{-1} x \iff \cos y = x$$

$$y = \arccos x \iff \cos y = x$$

Caused restrictions for $y = \cos^{-1} x$:

$$-1 \leq x \leq 1 \wedge 0 \leq y \leq \pi$$

15.3 Inverse Tangent Function

Restrictions for $y = \tan x$:

$$-\frac{\pi}{2} < x < \frac{\pi}{2}$$

Vertical Asymptotes for $y = \tan x$:

$$x = -\frac{\pi}{2}, \frac{\pi}{2}$$

Restrictions cause Horizontal Aymptotes for $y = \tan^{-1} x$:

$$y = -\frac{\pi}{2}, \frac{\pi}{2}$$

15.4 Inverse Secant Function

Restrictions for $y = \sec x$:

$$0 \leq x \leq \pi, x \neq \frac{\pi}{2}$$

Vertical Asymptotes for $y = \sec x$:

$$x = \frac{\pi}{2}$$

Restrictions cause Horizontal Asymptotes for $y = \sec^{-1} x$:

$$y = 0, y = \pi$$

15.5 Inverse Cosecant Function

Restrictions for $y = \csc x$:

$$-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}, x \neq 0$$

Vertical Asymptotes for $y = \csc x$: $x = 0$ Restrictions cause Horizontal Asymptotes for $y = \csc^{-1} x$:

$$y = -\frac{\pi}{2}, y = \frac{\pi}{2}$$

15.6 Inverse Cotangent Function

Restrictions for $y = \cot x$:

$$0 < x < \pi$$

Vertical Asymptotes for $y = \cot x$:

$$x = 0, x = \pi$$

Restrictions cause Horizontal Asymptotes for $y = \cot^{-1} x$:

$$y = 0, y = \pi$$

15.7 Inverse Function Use Cases

1. You can use the inverse of a trig function to determine the input to the original function based on a given output which does not provide a measure of the inputted angle but does provide the output.

$$\sin(\sin^{-1} 2\pi) = 2\pi$$

2. You can use values of a trig function to derive the values of a reference triangle which can then be used traditionally to determine an encapsulating function.

$$\tan\left[\cos^{-1}\left(\frac{\sqrt{x^2-9}}{x}\right)\right] = \frac{3}{x}$$

$$\left(\sqrt{x^2-9}\right)^2 + b^2 = x^2$$
$$x^2 - 9 + b^2 = x^2$$
$$-9 + b^2 = 0$$
$$b^2 = 9$$
$$b = 3$$
$$opp = 3 \wedge adj = \sqrt{x^2-9} \wedge hyp = x$$

16 5.1: Algebraic manipulation of Trigonometric Functions

16.1 LCD of Trigonometric Functions and Variables

$$LCD\ of\ \frac{a}{b}; a = b$$

When you apply the LCD:

$$\frac{a}{b} \rightarrow \frac{a}{b}$$
$$a \rightarrow \frac{ab}{b}$$

This can also be done with trig functions:

$$LCD\ of\ \frac{\sin x}{\cos x}; \cos x = \cos x$$

When you apply the LCD:

$$\frac{\sin x}{\cos x} \rightarrow \frac{\sin x}{\cos x}$$
$$\cos x \rightarrow \frac{\cos x}{\cos x}$$

16.2 Properties of Trigonometric Functions

They have similar properties to that of a variable, you can multiply them, factor them, etc...

16.3 Proving Trigonometric Identities

1. Work with one side of the equation and keep the other side in mind as your final goal
2. If the expression is a product or quotient of factors, consider the reciprocal and quotient identities.

3. If squared terms are present, look to see if terms can be grouped into one of the forms of a pythagorean identity
4. If an expression involves a negative argument, consider using the even or odd function identities.
5. Apply basic algebraic techniques like factoring, multiplying terms, combining like terms, and writing factions with a common denominator.
6. Consider writing expressions literally in terms of sine and cosine.
7. Make sure to remember that you can algebraically manipulate the identity expressions in order to suit your expressions needs.

17 5.2: Sum and Difference Formulas

17.1 Chord

A straight line segment whose endpoints both lie on the circumference of the circle. The diameter, which passes through the center of the circle, is the longest possible chord.

17.2 Sine Sum Formula

$$\sin(u+v) = \sin u \cos v + \cos u \sin v$$

17.3 Sine Difference Formula

$$\sin(u-v) = \sin u \cos v - \cos u \sin v$$

17.4 Cosine Sum Formula

$$\cos(u+v) = \cos u \cos v - \sin u \sin v$$

17.5 Cosine Difference Formula

$$\cos(u-v) = \cos u \cos v + \sin u \sin v$$

17.6 Tangent Sum Formula

$$\tan(u+v) = \frac{\tan u + \tan v}{1 - \tan u \tan v}$$

17.7 Tangent Difference Formula

$$\tan(u-v) = \frac{\tan u - \tan v}{1 + \tan u \tan v}$$

18 5.3: Double-Angle, Power-Reducing, and Half-Angle Measurements

18.1 Double-Angle Formulas

Since $\sin 2\theta = \sin(\theta + \theta)$, for angles which have coefficients applied to their arguments, we can simplify them using the sum and difference formulas for any of the respective trig. functions.

$$\sin 2\theta = \sin\theta\cos\theta + \cos\theta\sin\theta$$

The trig. functions have a variety of different identities which can be proven:

$$\sin 2\theta = 2\sin\theta\cos\theta$$

$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$

$$\cos 2\theta = 1 - 2\sin^2\theta$$

$$\cos 2\theta = 2\cos^2\theta - 1$$

$$\tan 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta}$$

18.2 Power-Reducing Formulas

$$\sin^2\theta = \frac{1 - \cos 2\theta}{2}$$

$$\cos^2\theta = \frac{1 + \cos 2\theta}{2}$$

$$\tan^2\theta = \frac{1 - \cos 2\theta}{1 + \cos 2\theta}$$

18.3 Half-Angle Formulas

Where the sign of a is determined by the quadrant in which the angle $\frac{a}{2}$ lies:

$$\sin\frac{a}{2} = \pm\sqrt{\frac{1 - \cos a}{2}}$$

$$\cos\frac{a}{2} = \pm\sqrt{\frac{1 + \cos a}{2}}$$

$$\tan\frac{a}{2} = \pm\sqrt{\frac{1 - \cos a}{1 + \cos a}} = \frac{1 - \cos a}{\sin a} = \frac{\sin a}{1 + \cos a}$$